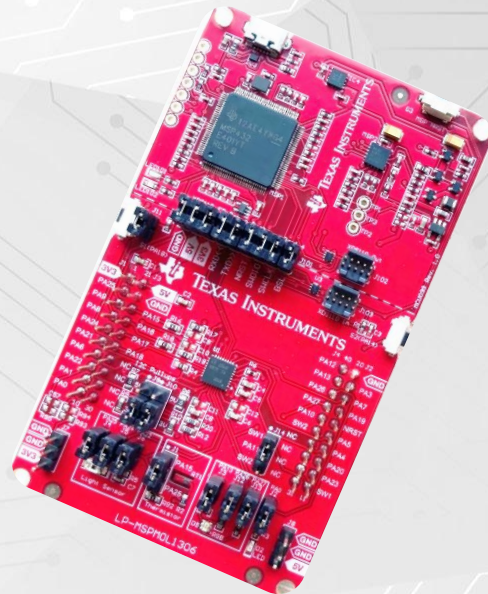


LP-MSPM0G3507快速入门

谢胜祥

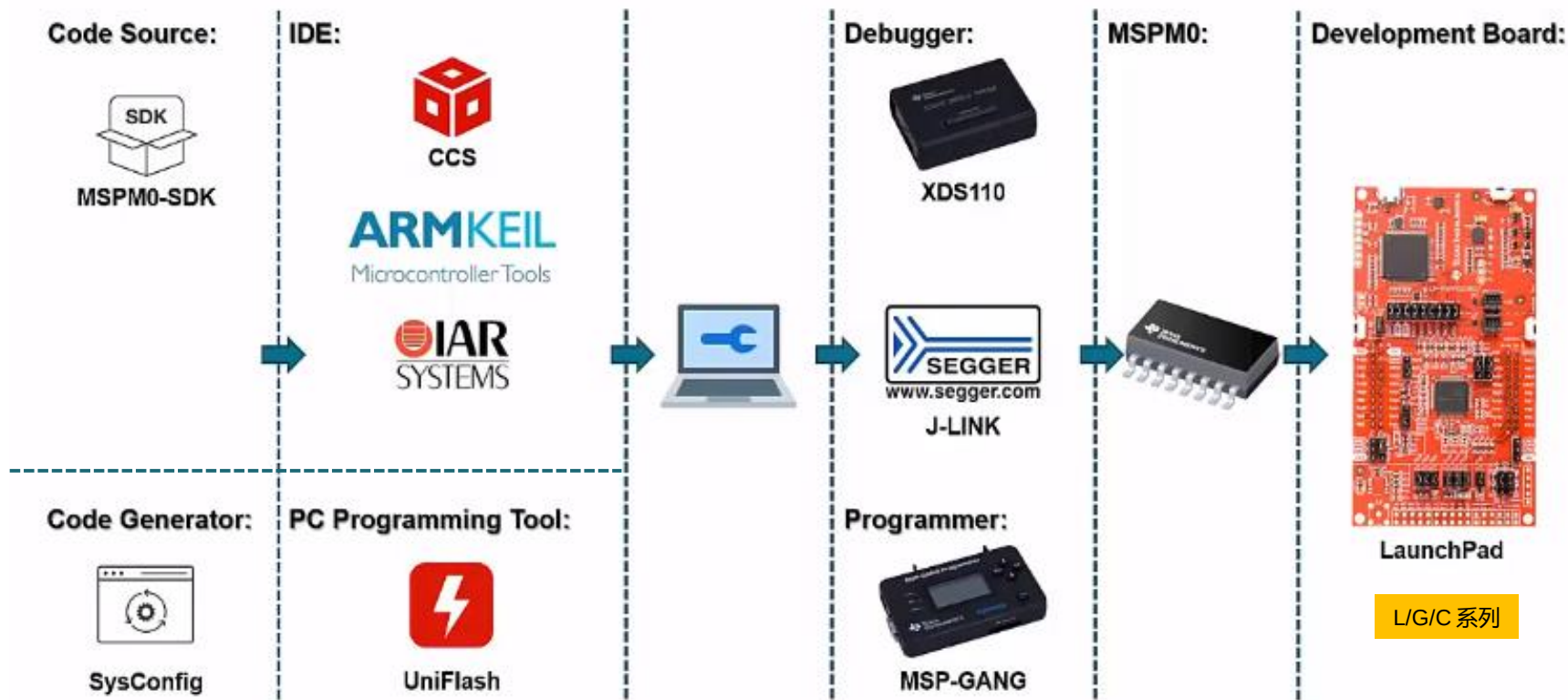
shengxiang-xie@ti.com



概述

- MSPM0 MCU的开发环境介绍
- MSPM0 MCU在教学和竞赛中的资料简介
- CCS + MSPM0 SDK的使用
- 例程操作与演示

MSPM0 MCU的开发环境



基本开发环境搭建

PC端的软件安装：

1. 集成开发环境 *(4选1)*
 - 1) Code Composer Studio
 - 2) CCS Theia
 - 3) Keil
 - 4) IAR for ARM
2. SysConfig图形配置工具
3. MSPM0 SDK软件开发包

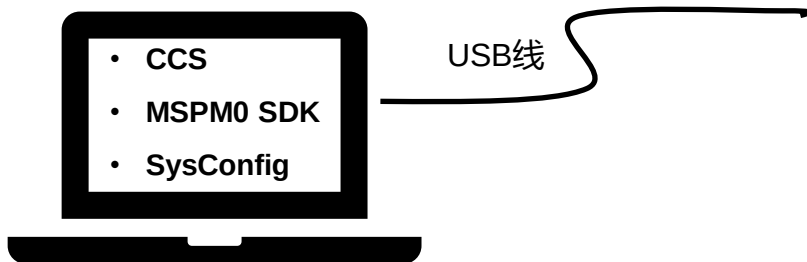
硬件平台选择：

1. TI Launchpad套件
 - 1) LP-MSPM0L1306
2. 第三方套件

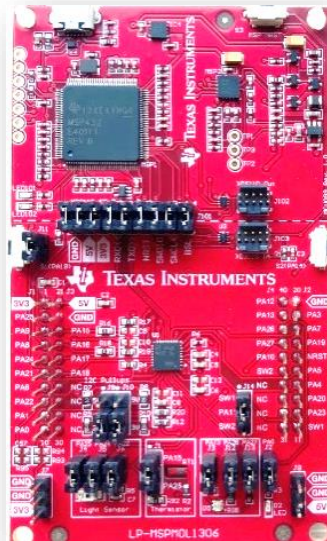
今日培训的开发环境与平台

基本目标

- CCS、SysConfig软件的操作
- 运行MSPM0 SDK例程



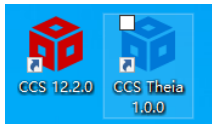
MSPM0 MCU Launchpad



软件的下载/安装

- CCS版本的区别

- CCS vs CCS Theia
- Desktop vs Web



- 下载完整安装包 vs 在线引导安装包

- 在线安装网络不稳定, 容易出错

- 安装路径不要包含中文等字符

- 学生喜欢用中文用户名的问题, 修改中文用户名/新建用户/重装系统

- 安装过程中产品的选择

- MSPM0 32bits arm cortex-M0+ ...

- MSPM0SDK、SysConfig的安装路径推荐

- 会自动关联搜索

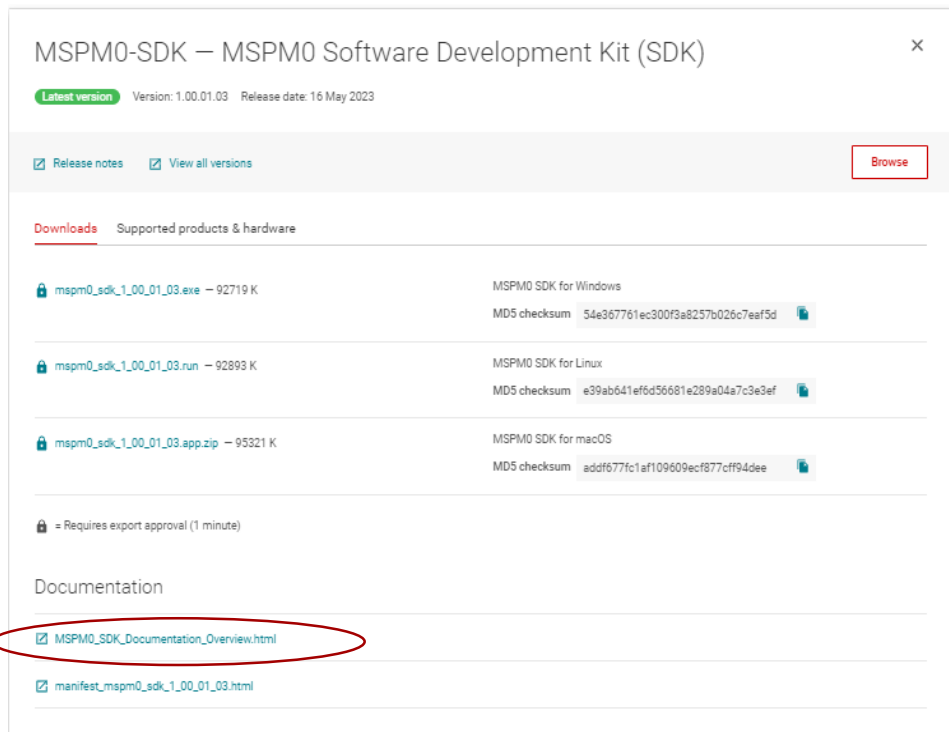
实践环节01. 安装软件及SDK包

完成以下2个软件的安装：

1. Code Composer Studio集成开发环境
2. MSPM0 SDK软件开发包

软件版本的依赖要求？

MSPM0 SDK文档查看



The screenshot shows the MSPM0-SDK webpage. At the top, it says "MSPM0-SDK — MSPM0 Software Development Kit (SDK)" with a close button. Below this, it indicates "Latest version" and "Version: 1.00.01.03" with a "Release date: 16 May 2023". There are links for "Release notes" and "View all versions", and a "Browse" button. The "Downloads" section lists three download options: "mspm0_sdk_1_00_01_03.exe" (92719 K) for Windows, "mspm0_sdk_1_00_01_03.run" (92893 K) for Linux, and "mspm0_sdk_1_00_01_03.app.zip" (95321 K) for macOS. Each option includes an MD5 checksum and a download icon. A note indicates that the download requires export approval (1 minute). The "Documentation" section lists two links: "MSPM0_SDK_Documentation_Overview.html" (circled in red) and "manifest_mspm0_sdk_1_00_01_03.html".



1. 打开MSPM0 SDK产品页面:
<https://www.ti.com/tool/MSPM0-SDK>
2. 点击 Download选项, 弹出选项框
3. 查看 Documentation
4. [点击 MSPM0_SDK_Documentation_Overview.html](#)

MSPM0 SDK概览

MSPM0 SDK 概述

MSPM0 软件开发套件 (SDK) 不仅提供了丰富全面的软件资源, 还提供了相应的工具和文档, 优化的性能和存储器利用率, 通过软件开发套件 (SDK) 中的各种驱动程序, 软件库和易于使用的示例提供一致的用户体验, 旨在打造简单易用的统一用户资源平台。

MSPM0 SDK Documentation

- Release Notes
- Quick Start Guides
- MSPM0 SDK User's Guide
- Manifest
- Early Samples Migration Guide

SDK Components

DriverLib Documentation

- DriverLib Guide

TI Drivers Documentation

- TI Drivers Overview

Middleware Documentation

- Boot Image Manager
- Brushed Motor Control Library
- Diagnostic Library
- EEPROM Emulation Library
- GUI Composer Library
- Hall Sensored Trap Motor Control Library
- IQMath Library
- LIN Library
- Sensorless FOC Motor Control Library
- SMBus Library
- PMBus Library
- Stepper Motor Control Library

3rd Party Documentation

- CMSIS DSP

MSPM0 Tools Documentation

- MSPM0 Tools Guide

MSPM0 SDK 文档

- 版本说明
- 快速上手指南
- MSPM0 SDK 用户指南
- 清单
- 早期样品移植指南

SDK 组件

DriverLib 文档

- DriverLib 指南

TI Drivers 文档

- TI Drivers 指南

中间件文档

- Boot Image 管理器
- 有刷电机控制库
- Diagnostic 诊断库
- EEPROM 仿真库
- GUI Composer 库
- Hall Sensored Trap Motor Control Library
- IQMath 库
- LIN 库
- Sensorless FOC Motor Control Library
- SMBus 库
- PMBus 库
- 步进电机控制库

3rd Party Documentation

- CMSIS DSP

MSPM0工具文档

- MSPM0工具指南



版本发布文档 v1_20_00_05

MSPM0 SDK Release Notes

Table of Contents

- Important Notes
- Introduction
- Documentation
- What's Included
- What's New
- Documentation
 - Operating Systems Support
 - Dependencies
 - Device Support
 - Known Issues
 - Upgrade and Compatibility Information
 - Previous Releases
 - Versioning
 - Technical Support and Product Updates

Important Notes

Operating Systems Support

- Windows 10
- Ubuntu 18.04
- macOS Monterey

Dependencies

This release was validated with the following components:

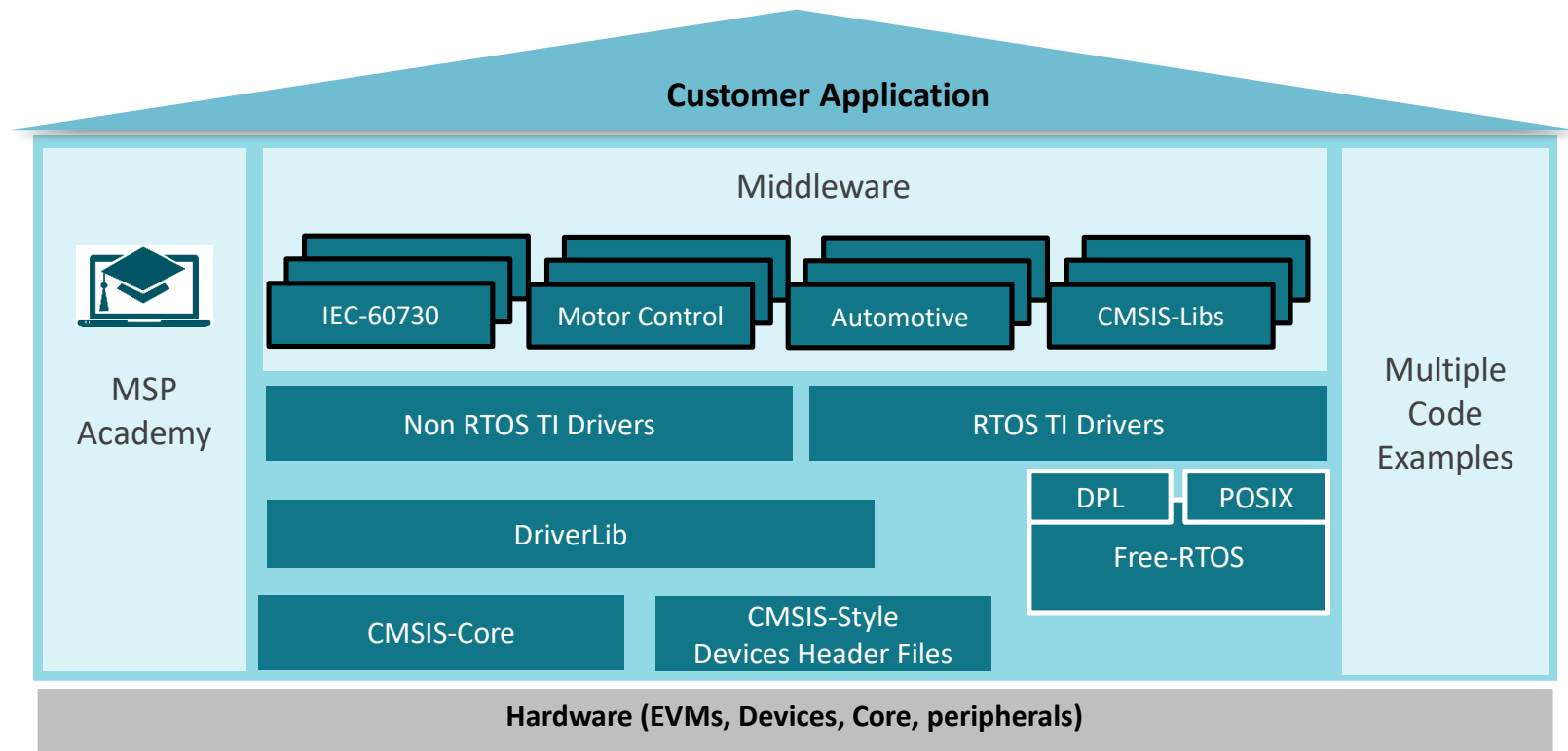
- Code Composer Studio
 - Code Composer Studio (CCS) Theia 1.1.0.
 - Code Composer Studio (CCS) 12.5.0.
- TI Arm Clang Compiler
 - TI Clang v3.2.0_LTS. Also included with CCS 12.5.0 and CCS Theia 1.2.
- SysConfig
 - SysConfig 1.18.x. Also included with CCS 12.5.0 and CCS Theia 1.2.
- GNU Code Generation Tools
 - Arm gcc-arm-none-eabi-9-2019-q4-major.
- IAR Embedded Workbench
 - IAR for Arm 9.40.2.
- Keil uVision
 - 5.38, with Arm Clang compiler v6.16 or later.

Device Support

- MSPM0L11xx/L13xx: Tested on production silicon.
- MSPM0G1x0x/G3x0x: Tested on production silicon.
- MSPM0C110x: Preliminary support.
- MSPS003FX: Preliminary support.



SDK软件开发包概述



外设驱动库DriverLib

MSPM0 SDK 概述

MSPM0 软件开发套件 (SDK) 不仅提供了丰富全面的软件资源, 还提供了相应的工具和文档, 优化的性能和存储器利用率, 通过软件开发套件 (SDK) 中的各种驱动程序, 软件库和易于使用的示例提供一致的用户体验, 旨在打造简单易用的统一用户资源平台。

MSPM0 SDK 文档

- 版本说明
- 快速上手指南
- MSPM0 SDK 用户指南
- 清单
- 早期样品移植指南

SDK 组件

DriverLib 文档

- DriverLib 指南

TI Drivers 文档

- TI Drivers 指南

中间件文档

- Boot Image 管理器
- 有刷电机控制库
- Diagnostic 诊断库
- EEPROM 仿真库
- GUI Composer 库
- Hall Sensored Trap Motor Control Library
- IQMath 库
- LIN 库
- Sensorless FOC Motor Control Library
- SMBus 库
- PMBus 库
- 步进电机控制库

3rd Party Documentation

- CMSIS DSP

MSPM0 工具文档

- MSPM0 工具指南

MSPM0 驱动程序库概述

引言

德州仪器 (TI) MSPM0 驱动程序库 (DriverLib) 是一组 功能齐全的 API, 用于配置、控制和操作 MSPM0 平台的硬件外设。

除了能够控制所有 MSPM0 外设, DriverLib 还使用户能够使用常见 ARM 外设, 例如中断 控制器 (NVIC) 和 SysTick。

DriverLib 旨在为程序员提供一个软件层, 与直接访问寄存器相比, 这样便于实现更高层次的编程。

使用 DriverLib API 可以配置和驱动 MSPM0 器件的几乎 每一个方面。通过使用 DriverLib 提供的高级软件 API, 用户可以创建功能强大且易于理解的代码, 这些代码可以在不同器件之间轻松地 移植。

API 指南

- MSPM0G11XX_G3X0X DriverLib 指南
- MSPM0L11XX_L13XX DriverLib 指南
- MSPM0C110X DriverLib 指南



 **Technology for Innovators™**

MSPM0L11XX_L13XX Driver Library 1.20.00.05

MSPM0 Driver Library

Introduction

The Texas Instruments MSPM0 Driver Library (DriverLib) is a set of fully functional APIs used to configure, control, and n

In addition to being able to control all MSPM0 peripherals, DriverLib also gives the user the ability to use common Arm

DriverLib for MSPM0 Series has been tested and compiled under a variety of different toolchains.

Toolchain
Texas Instruments Code Composer Studio (CCS) Theia 1.1.0
Texas Instruments Code Composer Studio (CCS) 12.5
IAR Embedded Workbench for Arm 9.40.2
GNU C Compiler 9.2.1 (gcc)
Keil uVision 5.38

Driverlib 模块概览

Modules

Here is a list of all modules:

- Analog to Digital Converter (ADC12)
- Advanced Encryption Standard (AES)
- DriverLib Common APIs
- Comparator (COMP)
- Cyclic Redundancy Check (CRC)
- Digital to Analog Converter (DAC12)
- Direct Memory Access (DMA)
- Flash Controller (FLASHCTL)
- General Purpose Amplifier (GPAMP)
- General Purpose Input/Output (GPIO)
- Inter-Integrated Circuit (I2C)
- Math Accelerator (MATHACL)
- Modular Controller Area Network (MCAN)
- Op-Amp (OPA)
- Real Time Clock (RTC)
- Real Time Clock (RTC_Common)
- Serial Peripheral Interface (SPI)
- Common General Purpose Timer (TIMx)
- General Purpose Timer Advance (TIMA)
- General Purpose Timer G (TIMG)
- True Random Number Generator (TRNG)
- Universal Asynchronous Receiver-Transmitter (UART)
- UART Extend (UART-Extend)
- UART Main (UART-Main)
- Voltage Reference (VREF)
- Window Watchdog Timer (WWDG)
- M0P Core
- Factory Region (FACTORYREGION)
- M0P Interrupts
- M0P System Tick Timer (SysTick)
- MSPM0G1X0X_G3X0X System Control (SYSCTL)
- RTC_Common

Functions

- | | | |
|----------------------|---|--|
| __STATIC_INLINE void | DL_GPIO_enablePower (GPIO_Regs *gpio) | Enables power on GPIO module. More... |
| __STATIC_INLINE void | DL_GPIO_disablePower (GPIO_Regs *gpio) | Disables power on gpio module. More... |
| __STATIC_INLINE bool | DL_GPIO_isPowerEnabled (GPIO_Regs *gpio) | Returns if power on gpio module. More... |
| __STATIC_INLINE void | DL_GPIO_reset (GPIO_Regs *gpio) | Resets gpio peripheral. More... |
| __STATIC_INLINE bool | DL_GPIO_isReset (GPIO_Regs *gpio) | Returns if gpio peripheral was reset. More... |
| __STATIC_INLINE void | DL_GPIO_initDigitalOutput (uint32_t pinIdx) | Configures a pin as a basic GPIO output. More... |
| __STATIC_INLINE void | DL_GPIO_initDigitalOutputFeatures (uint32_t pinIdx) | Configures a pin as a GPIO output. More... |
| __STATIC_INLINE void | DL_GPIO_initDigitalInput (uint32_t pinIdx) | Configures a pin as a basic GPIO input. More... |
| __STATIC_INLINE void | DL_GPIO_initDigitalInputFeatures (uint32_t pinIdx) | Configures a pin as a GPIO input. More... |
| __STATIC_INLINE void | DL_GPIO_initPeripheralFunction (uint32_t pinIdx) | Configure a pin to operate with peripheral functions. More... |
| __STATIC_INLINE void | DL_GPIO_initPeripheralOutputFunction (uint32_t pinIdx) | Configure a pin to operate with peripheral output functions. More... |



§ DL_GPIO_setPins()

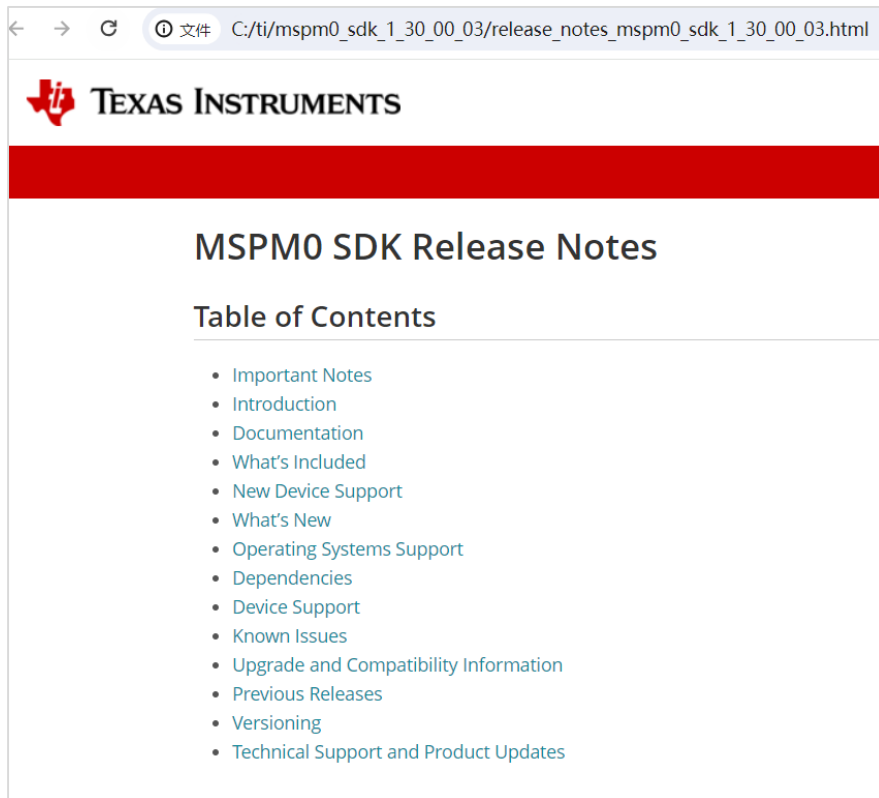
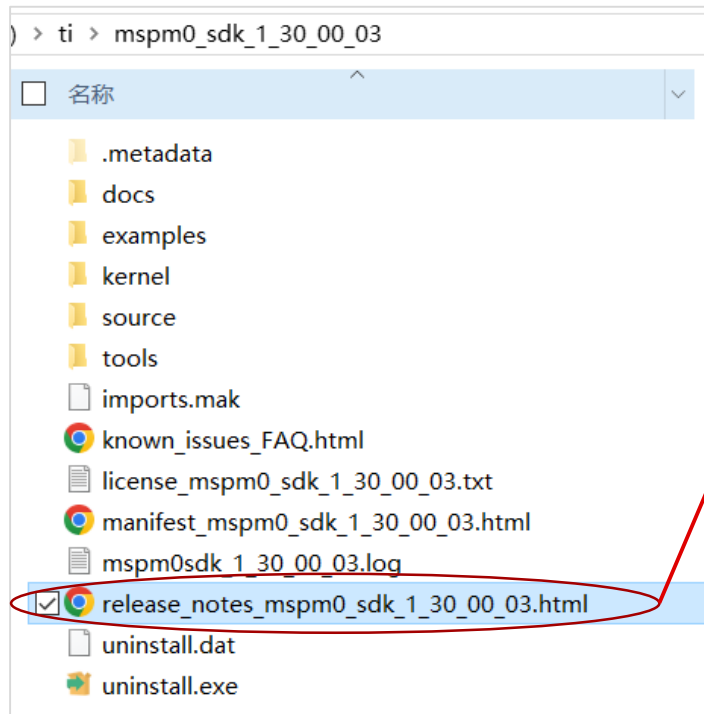
```
__STATIC_INLINE void DL_GPIO_setPins ( GPIO_Regs * gpio,  
                                       uint32_t pins  
                                       )
```

Set a group of GPIO pins.

Parameters

- [in] **gpio** Pointer to the register overlay for the peripheral
- [in] **pins** Pins to set high. Bitwise OR of **DL_GPIO_PIN**.

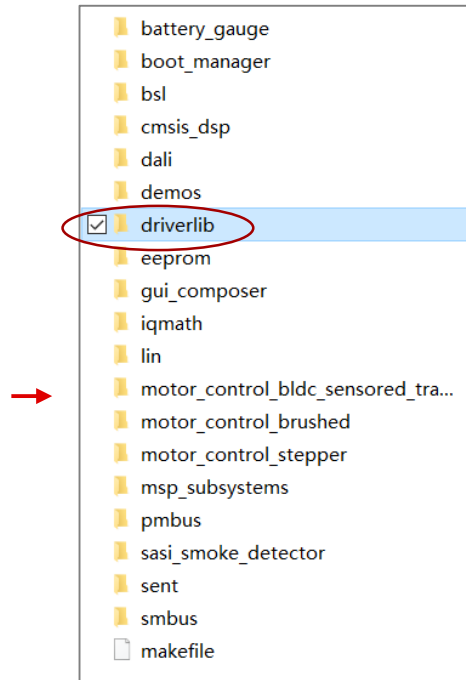
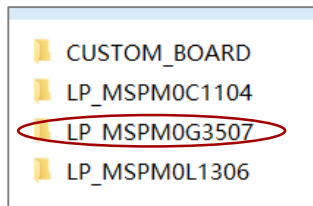
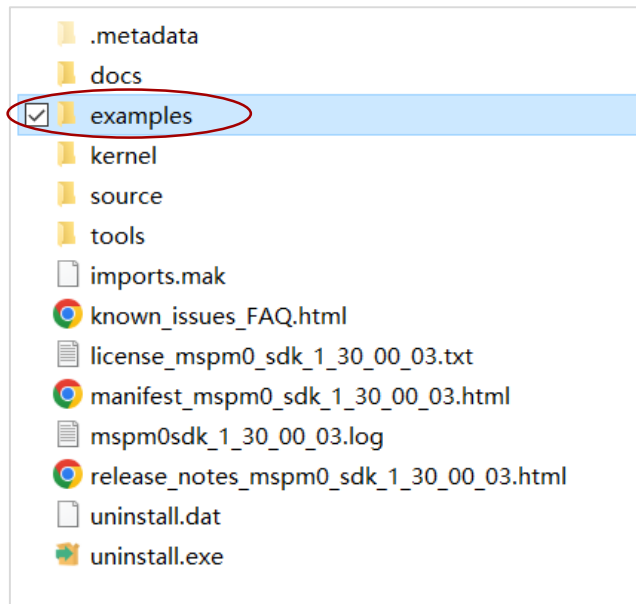
MSPM0 SDK本地说明文档



例程说明

MSPM0 SDK的安装目录

- C:\ti\mspm0_sdk_1_30_00_03\examples\nortos\LP_MSPM0G3507\driverlib



Driverlib例程

• C:\ti\mspm0_sdk_1_30_00_03\examples\nortos\LP_MSPM0G3507\driverlib

- adc12 14bit resolution
- adc12 internal temp sensor mathacl
- adc12 max freq dma
- adc12 max freq dma 8bit
- adc12 monitor supply
- adc12 sequence conversion
- adc12 simultaneous trigger event
- adc12 simultaneous trigger event stop
- adc12 single conversion
- adc12 single conversion vref external
- adc12 single conversion vref internal
- adc12 triggered by timer event
- adc12 triggered by timer event stop
- adc12 window comparator
- aes cbc 256 enc dec
- aes cfb 256 decrypt
- aes ofb 128 encrypt
- cinit bypass
- comp analog filter
- comp dac to timer event
- comp hs dac vref external
- comp hs tima pwm fault
- comp lp dac vref internal
- crc calculate checksum
- crc calculate checksum dma
- dac12 dma sampletimegen
- dac12 fifo sampletimegen
- dac12 fifo timer event
- dac12 fixed voltage vref internal
- dma block transfer
- dma fill data
- dma table transfer
- empty
- empty cpp
- empty driverlib src
- empty library
- empty non sysconfig
- event input triggers output
- flashctl blank verify
- flashctl dynamic memory protection
- flashctl ecc error injection
- flashctl multiple size read verify
- flashctl multiple size write
- flashctl nonmain memory write
- flashctl program with ecc
- gpamp buffer to adc
- gpamp general purpose rri
- gpio input capture
- gpio simultaneous interrupts
- gpio software poll
- gpio toggle output
- gpio toggle output cpp
- gpio toggle output hiz
- ☒ i2c controller rw multibyte fifo interrupts
- i2c controller rw multibyte fifo poll
- i2c controller target dynamic switching
- i2c multicontroller arbitration
- i2c target rw multibyte fifo interrupts
- i2c target rw multibyte fifo interrupts stop
- i2c target rw multibyte fifo poll
- mathacl mpy div op
- mathacl trig op
- mcan loopback
- mcan message rx
- mcan message rx tcant14x
- mcan multi message tx
- mcan multi message tx tcant14x
- mcan single message tx
- nvic interrupt disable
- nvic interrupt grouping
- opa burnout current source to adc
- opa dac8 output buffer
- opa general purpose rri
- opa inverting pga with dac
- opa non inverting pga
- opa signal chain to adc
- rtc calendar alarm standby
- rtc offset calibration lfx
- rtc periodic alarm lfosc standby
- rtc periodic alarm lfx standby
- spi controller command data control
- spi controller echo interrupts
- spi controller fifo dma interrupts
- spi controller internal loopback poll
- spi controller multibyte fifo poll
- spi controller register format
- spi controller repeated fifo dma interrupts
- spi peripheral echo interrupts
- spi peripheral fifo dma interrupts
- spi peripheral multibyte fifo poll
- spi peripheral register format
- spi peripheral repeated fifo dma interrupts
- sram parity
- sysctl frequency clock counter
- sysctl hfx run
- sysctl lfx standby
- sysctl mclk syspll
- sysctl power policy sleep to standby
- sysctl power policy sleep to stop
- sysctl shutdown
- systick periodic timer
- tima timer mode periodic repeat count
- tima timer mode pwm dead band
- tima trigger fail mechanism
- timg 32bit timer mode periodic sleep
- timg 32bit timer mode pwm edge sleep
- timg qei mode
- timx timer mode capture duty and period
- timx timer mode capture edge capture
- timx timer mode compare edge count
- timx timer mode one shot standby
- timx timer mode periodic sleep
- timx timer mode periodic standby
- timx timer mode periodic stop
- timx timer mode pwm center stop
- timx timer mode pwm cross trigger stop restore
- timx timer mode pwm edge sleep
- timx timer mode pwm edge sleep shadow
- trng sample
- trng sample stop restore
- uart echo interrupts standby
- uart extend irda receive packet
- uart extend irda send packet
- uart extend manchester echo
- uart extend manchester send packet
- uart external loopback interrupt
- uart internal loopback standby restore
- uart rs485 receive packet
- uart rs485 send packet
- uart rw multibyte fifo poll
- uart rx hw flow control
- uart rx multibyte fifo dma interrupts

138↑

参考资料 – 芯片产品页面

 **TEXAS INSTRUMENTS**

搜索

交叉参考搜索

• 数据手册

• 用户指南

• 勘误表

• 技术文档

简体中文 | 中国 | CNY

产品 应用 设计和开发 质量和可靠性 订购资源 关于 TI

购物车

通知

立即订购

主页 / 微控制器 (MCU) 和处理器 / 基于 Arm 的微控制器 / Arm Cortex-M0+ MCU

新

MSPM0G3507  正在供货

具有 128KB 闪存、32KB SRAM、2 个 12 位 4Msps ADC、DAC、3 个 COMP、3 个运算放大器、CAN-FD、MATHACL 的 80MHz Arm M0+ MCU

数据表  MSPM0G350x 具有 CAN-FD 接口的混合信号微控制器 数据表 (Rev. A) PDF | HTML

 最新英语版本 (Rev.B) PDF | HTML

用户指南  MSPM0 G-Series 80-MHz Microcontrollers Technical Reference Manual (Rev. A) (英语)

勘误  MSPM0G350x, MSPM0G150x Mixed-Signal Microcontrollers Errata (Rev. A) (英语)



寄存器的说明 – 用户指南

≡

MSPM0 L-Series 32-MHz Microcontrollers Technical Reference Manual (Rev. C)

410 / 1119

– 100% +

🔍

📁

5 NVM (Flash)

📁

6 Events

📁

7 IOMUX

7.1 IOMUX Overview

7.2 IOMUX Operation

7.3 IOMUX (PINCMx) Register Format

7.4 IOMUX Registers

📁

8 GPIO

8.1 GPIO Overview

8.2 GPIO Operation

8.3 GPIO Registers

📁

9 ADC

9.1 ADC Overview

9.2 ADC Operation

9.3 ADC12 Registers

📁

10 COMP

📁

11 OPA

📁

12 GPAMP

📁

13 VREF

📁

14 UART

📁

15 SPI

TEXAS
INSTRUMENTS

www.ti.com

GPIO

8.3.14 ISET (Offset = 1040h) [Reset = 00000000h]

ISET is shown in [Figure 8-17](#) and described in [Table 8-17](#).

Return to the [Summary Table](#).

Interrupt set. Allows interrupts to be set by software (useful in diagnostics and safety checks). Writing a 1 to a bit in ISET will set the event and therefore the related RIS bit also gets set. If the interrupt is enabled through the mask, then the corresponding MIS bit is also set.

Figure 8-17. ISET

31	30	29	28	27	26	25	24
DIO31	DIO30	DIO29	DIO28	DIO27	DIO26	DIO25	DIO24
W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h
23	22	21	20	19	18	17	16
DIO23	DIO22	DIO21	DIO20	DIO19	DIO18	DIO17	DIO16
W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h
15	14	13	12	11	10	9	8
DIO15	DIO14	DIO13	DIO12	DIO11	DIO10	DIO9	DIO8
W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h
7	6	5	4	3	2	1	0
DIO7	DIO6	DIO5	DIO4	DIO3	DIO2	DIO1	DIO0
W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h	W-0h

Table 8-17. ISET Field Descriptions

Bit	Field	Type	Reset	Description
31	DIO31	W	0h	DIO31 event 0h = No effect 1h = Sets DIO31 in RIS register
30	DIO30	W	0h	DIO30 event 0h = No effect 1h = Sets DIO30 in RIS register
29	DIO29	W	0h	DIO29 event 0h = No effect 1h = Sets DIO29 in RIS register

22

 TEXAS INSTRUMENTS

参考资料 – LP-MSPM0G3507产品页面

LP-MSPM0G3507

适用于 80MHz Arm® Cortex®-M0+ MCU 的 MSPM0G3507
LaunchPad™ 开发套件

立即订购

概述 | 立即订购并开发 | 技术文档 | 相关设计资源 | 支持与培训

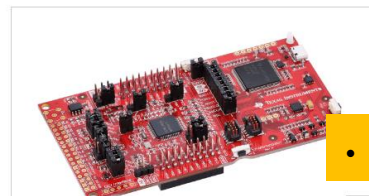
概述

描述和特性 支持的产品

LP-MSPM0G3507 LaunchPad™ 开发套件是基于 MSPM0G3507 的易用型评估模块 (EVM)，包含在 MSPM0G3507 M0+ MCU 平台上开始开发所需要的全部资源，包括用于编程、调试和能量测量的板载调试探针。该板包含三个按钮、两个 LED（其中一个为 RGB LED）以及模拟温度传感器和光传感器，还具有一个外部缓冲器，用于显示 4MSPS 时的高速 ADC 性能。

特性

- 外部环境光传感器
- RGB 显示 PWM 控制
- 用于显示 4MHz 高速 ADC 的外
- 外部温度传感器



• EVM用户指南

技术文档

类型

全部



标题

按关键字筛选标题



下载最新的英文版本

日期 ↓↑

EVM 用户指南

MSPM0G3507 LaunchPad 开发套件 (LP-MSPM0G3507) (Rev. C)

PDF | HTML 英语版 (Rev.C)

PDF | HTML 2024年 1月 11日



参考资料—培训视频

MSPM0 training series

这个系列将包含MSPM0的入门教学指引，如开发生态的组成说明，片上外设的特点介绍，特色应用场景的系统分享等等。旨在帮助大家更快的入门MSPM0，减小学习难度，降低开发门槛。



MSPM0 I/O module introduction

GPIO module

Key Features

- Standby-drive with wake IO is for SNU/IDOWN wake-up
- Up to 5 high-speed GPIOs (HSIO) support up to 60MHz frequency
- Up to 4 high-drive GPIOs (HIO) are able to output 20mA current to drive the load
- 5V tolerant open-drain GPIOs enable 5V communication without level shifter devices
- Unused pin should be set as GPIO output internal pull-down
- 0 wait state MMR access from CPU enhance SPI or I2C simulation using GPIO

Key Differences between G and L MCUs

- There is no high-drive IO on MSPM0Lx MCUs
- NRESET can be used as GPIO on 20-pin and 16-pin packages on MSPM0Lx MCUs

Digital IO Type can be found on device data sheet

IO Structure	Low-power I/O	Drive Strength	High-speed I/O	Pull-up/Pull-down	Push/Pull	Wake-up Logic
Standard-drive	Y			Y	Y	Partial
High-drive	Y	Y		Y	Y	Y
High-speed	Y	Y		Y	Y	Y
Open-drain	Y		Y		Y	Y

MSPM0 I/O (IOMUX) Introduction

00:07:17 | 2023 年 5 月 11 日

关于 文字记录

这个视频中将着重介绍MSPM0上IOMUX的结构与特点，同时也会给出相关的入门资源。

资源

↓ 演示

→ Learn more

• 支持倍速

• PPT下载

• 系列视频

主题

全部展开

MSPM0 ecosystem training series (10) ▼

MSPM0 peripheral training series (22) ▲

参考资料 – MSPM0 academy



Resource Explorer

Filter by keyword

Arm®-based microcontrollers

Device documentation

Embedded Software

Hardware kits and boards

Training

AM24x Academy - 2.00.00.01

AM26x Academy - 2.00.00.01

AM27x Academy - 2.00.00.00

MSPM0 Academy - 2.00.00.00

MSPM0 Academy 教程 - 2.00.00.00

Analog Peripherals

CPU Core and Memory

Serial Communication

MSP 学院 – I2C 简介实验室

MSP 学院 – SPI 简介实验室

UART 简介实验室

Timers

SimpleLink Academy for MSP432E4 - 3.30.01.00

Arm®-based processors

C2000™ real-time microcontrollers

Development Tools

mmWave radar sensors

MSP430™ microcontrollers

Wireless connectivity

MSPM0 Academy 教程

欢迎来到MSP教程，这个教程面向全部MSPM0开发者提供了大量实际应用的解决方案。本MSP教程基于MSPM0 MCU 针对大量不同的应用场景提供了简单上手的教程。

Search MSPM0 Academy 教程

Analog Peripherals

ADC 简介实验室

本文主要介绍基于MSPM0芯片的ADC外设的基础知识及参考示例代码。

MSP 学院 – OpAmp 简介实验室

MSPM0 OPA（运算放大器）模块介绍

VREF 简介实验室

MSPM0 MCU 上的 VREF 模拟电压参考简介。



参考资料 -- MSPM0 MCU 快速参考指南

三 MSPM0 MCU 快速参考指南 (Rev. A) 11 / 14 148%

内容

商标

1 MSPM0 概述

- 1.1 产品系列
- 1.2 生态系统

2 MSPM0 设计资源

- 2.1 培训资源
 - 2.1.1 MSP Academy
 - 2.1.2 高精度实验室
- 2.2 开发工具
- 2.3 嵌入式软件资源
 - 2.3.1 MSPM0 软件开发套件 (SDK)
 - 2.3.2 实时操作系统 (RTOS)
 - 2.3.3 子系统示例
- 2.4 硬件工具和 EVM

3 MSPM0 应用资源

- 3.1 应用手册
- 3.2 应用特定的资源

4 首字母缩写词及定义

5 修订历史记录

3.1 应用手册

以下应用手册对 MSPM0 MCU 和外设进行了介绍, 可供您在开发设计时参考。

- 迁移指南
 - STM32 至 MSPM0 迁移指南
 - MSP430 至 MSPM0 迁移指南
- MSPM0 子系统
 - PWM LED 驱动器
 - 可编程增益放大器
 - 跨阻放大器
 - 5V 接口
 - 采用乒乓模式 DMA 的 ADC
 - 热敏电阻温度传感器
- 硬件
 - MSP 器件故障排除指南
 - MSP 32kHz 晶体振荡器
 - MSPM0Gx MCU 硬件开发指南
 - MSPM0Lx MCU 硬件开发指南
- 模拟
 - 借助 MSPM0 精密模拟, 轻松进行系统设计
 - 《模数规格和性能特性术语表》
 - 《对 MSP ADC 进行一般过采样以提高分辨率》
 - 《高速模数转换器基础知识》
 - 《使用 ADC12_A 实现热电偶接口》



实践环节02. 熟悉CCS的基本操作

1. 熟悉CCS的操作界面
2. 工程的基本操作（新建/导入/复制/构建等）
3. 在线资源管理器介绍

CCS启动界面



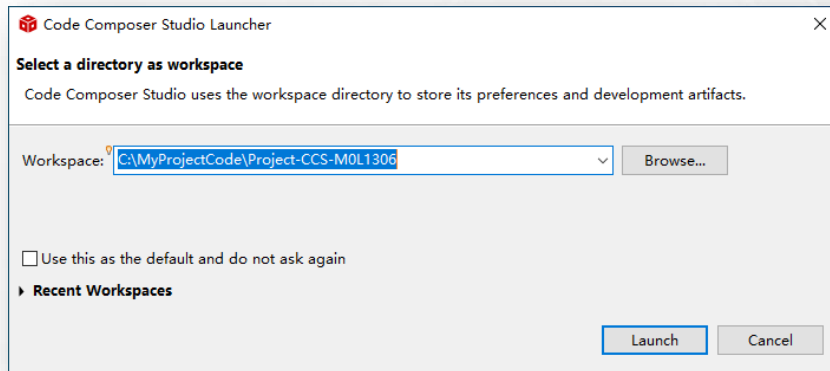
双击快捷图标，启动CCS；

“启动界面”如下图所示



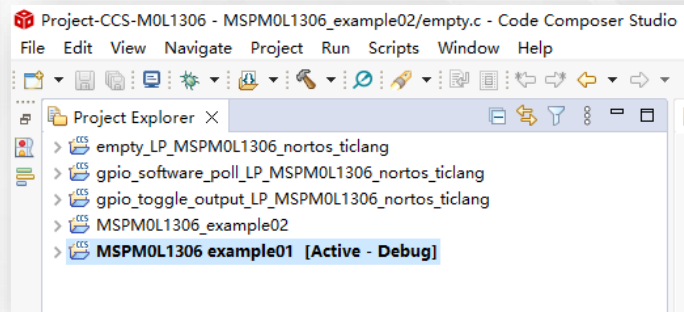
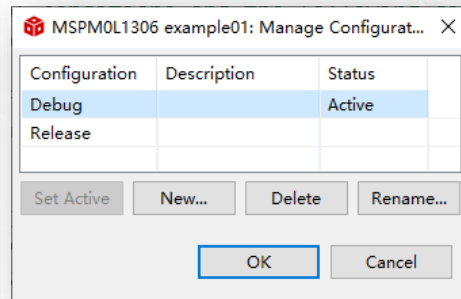
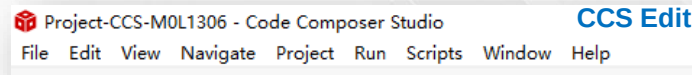
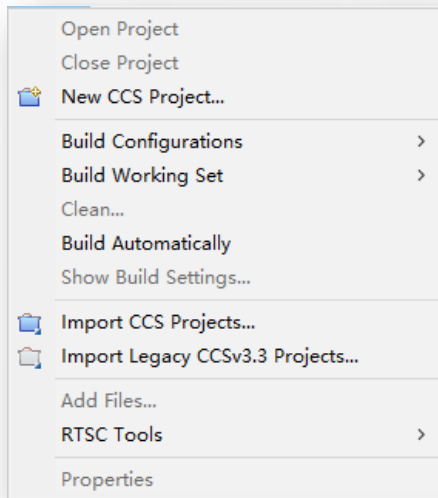
- **选择工作空间的存放路径：**

- 不能出现中文路径，有可能会导致闪退；
- 可根据需要，勾选是否用作默认路径；
 - 若勾选上，下次就不出现该选项框

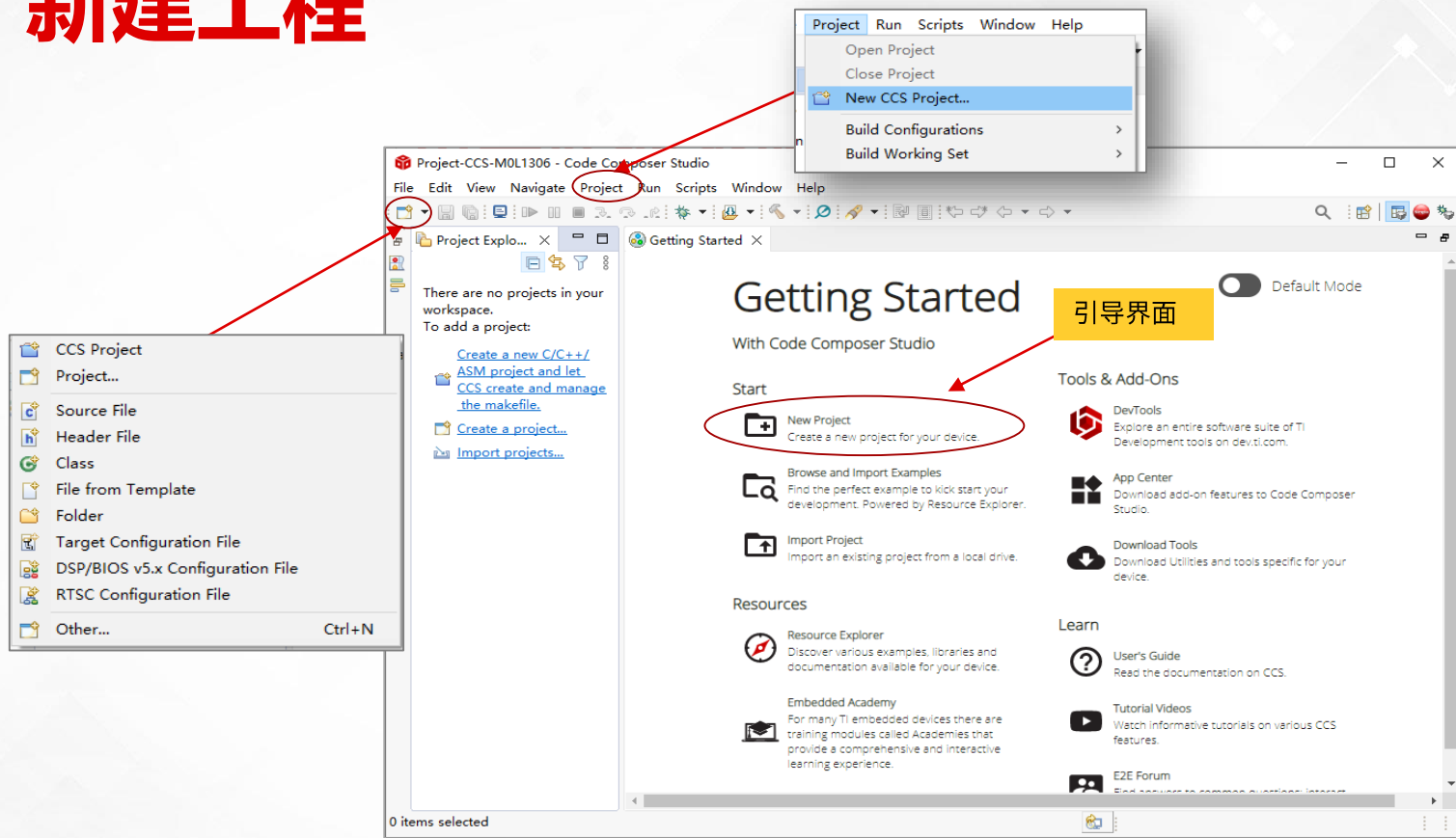


工程的基本操作

- 打开工程(Open Project)
- 关闭工程(Close Project)
- 新建工程(New CCS Project...)
- 导入工程(Import CCS Projects...)
- 工程属性设置Properties
 - 一般右键再选择比较快



新建工程



新建工程的配置

New CCS Project

① Project name must be specified

Target: <select or type filter text> Generic Arm7 Device

Connection: Verify...

Arm7 [Arm]

Project name:

☒ Use default location

Location: C:\MyProjectCode\Project-CCS-M0L1306 Browse...

Compiler version: TI v20.2.7.LTS More...

Project type and tool-chain

Project templates and examples

type filter text

- Empty Projects
 - Empty Project
 - Empty Project (with main.c)
 - Empty Assembly-only Project
 - Empty RTSC Project
- Basic Examples
 - Hello World

Creates an empty project initialized for the selected device. The project will contain an empty 'main.c' source-file.

Open [Resource Explorer](#) to browse a wide selection of example projects...

Open [Import Wizard](#) to find local example projects for selected device...

< Back Next > Finish Cancel

选择产品类型和型号

调试器类型

- 选择型号后会自动关联出来供选择

工程的名称，尽量不要用特殊字符

这两个可以展开1个

选择创建的工程模板

New CCS Project

Create a new CCS Project.

Target: MSPM0 MSPM0L1306

Connection: Texas Instruments XDS110 USB Debug Probe [Default] Verify...

Cortex M [Arm]

Project name: MSPM0L1306 example01

☒ Use default location

Location: C:\MyProjectCode\Project-CCS-M0L1306\MSPM0L1306 exe Browse...

Compiler version: TI Clang v2.1.2.LTS More...

Project type and tool-chain

Output type: Executable

Output format: eabi (ELF)

Device endianness: little

Linker command file: Browse...

Runtime support library: Browse...

Project templates and examples

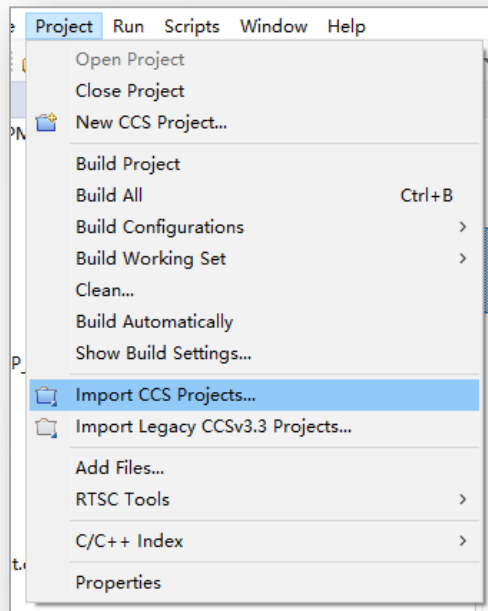
Open [Resource Explorer](#) to browse a wide selection of example projects...

Open [Import Wizard](#) to find local example projects for selected device...

< Back Next > Finish Cancel

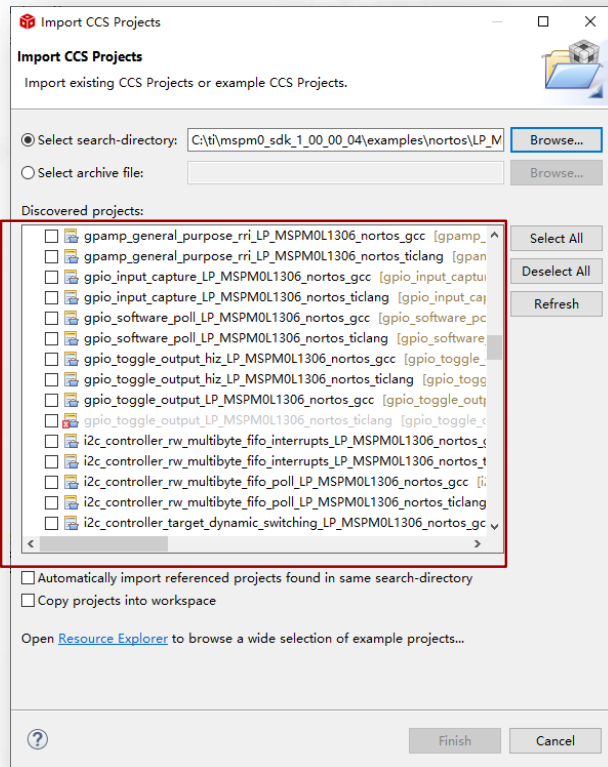
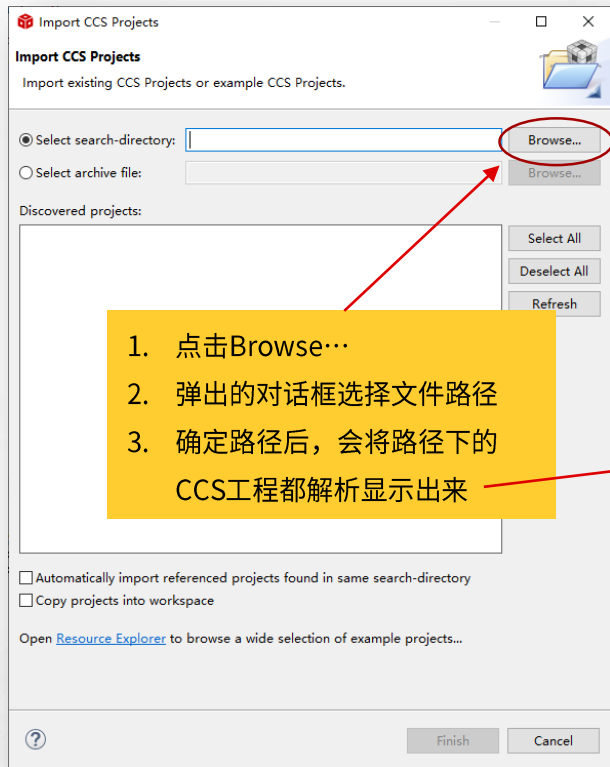
本地导入工程等操作

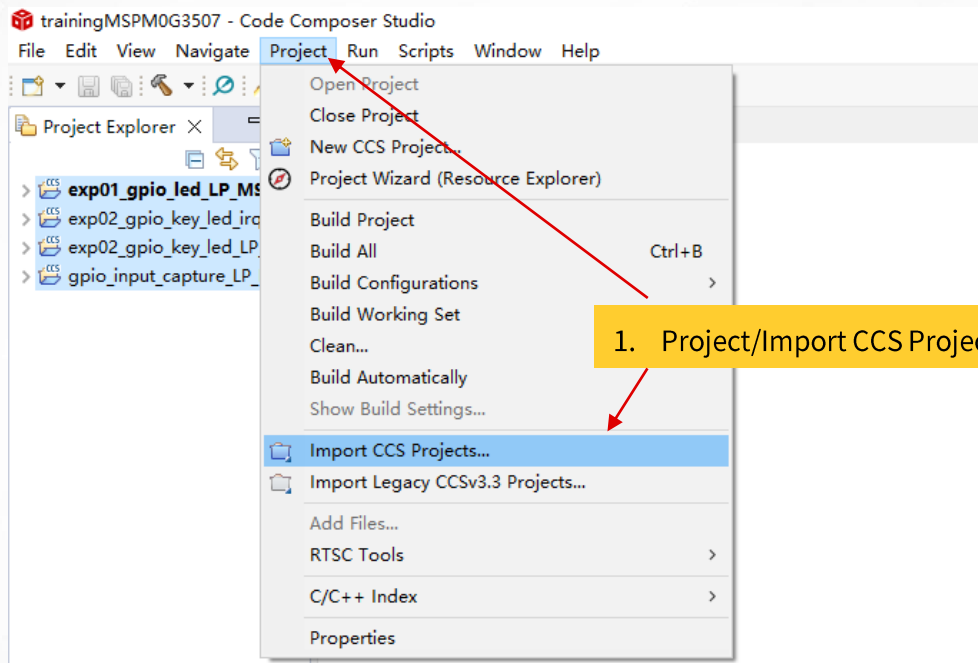
导入工程



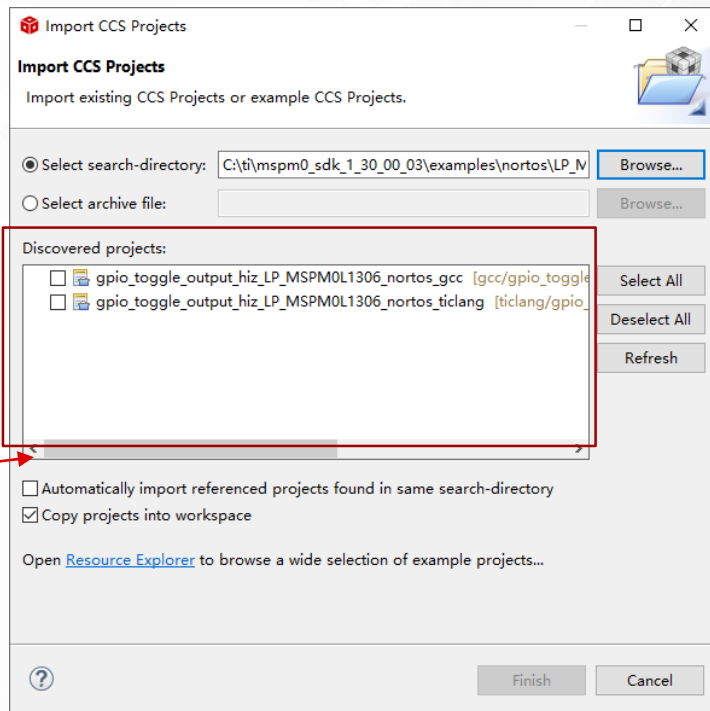
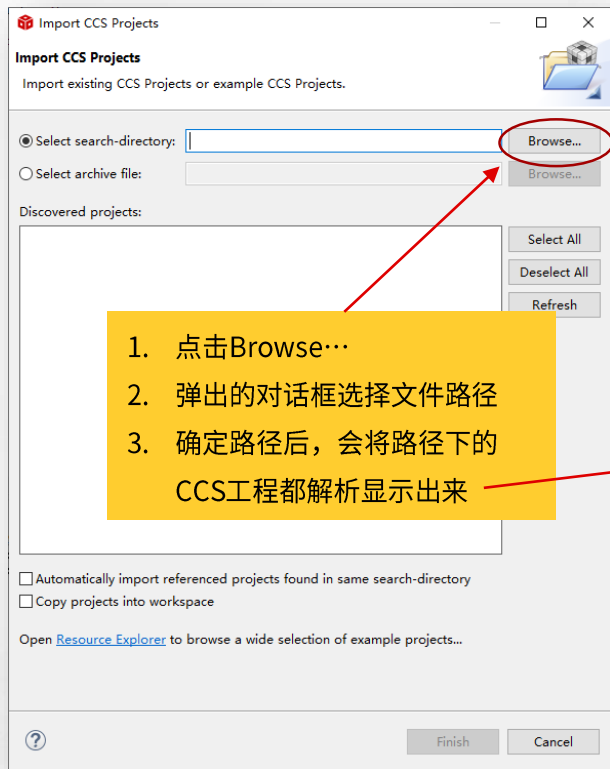
- 可以通过资源管理器的导入工程（图像化的展示界面）；
 - 在线资源管理器
 - 本地资源管理器（离线）
- 可以导入本地的工程（菜单操作界面）
 - eg. SDK里的工程
 - 用户自己创建的工程

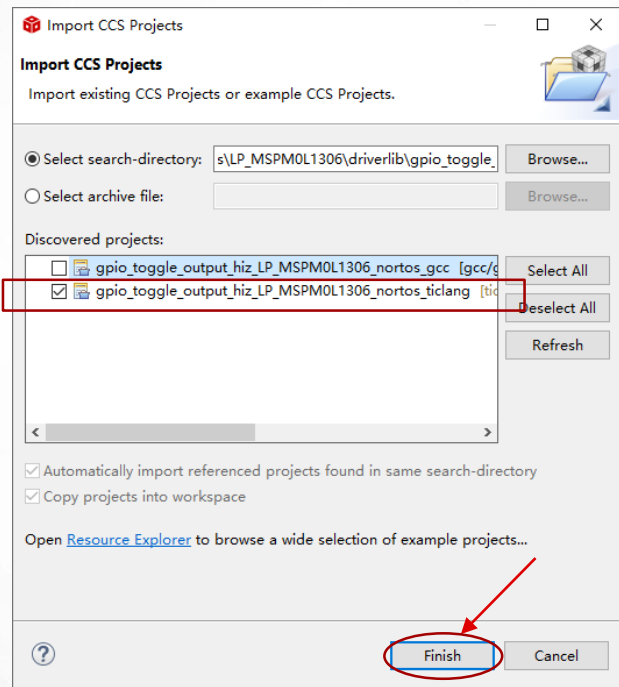
导入工程示例



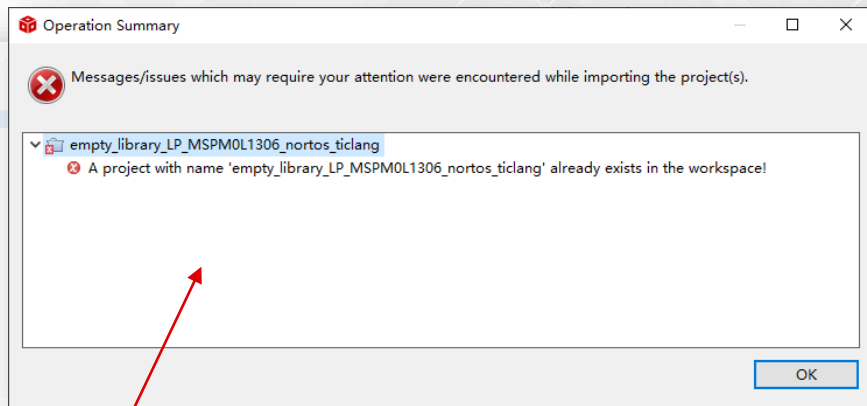
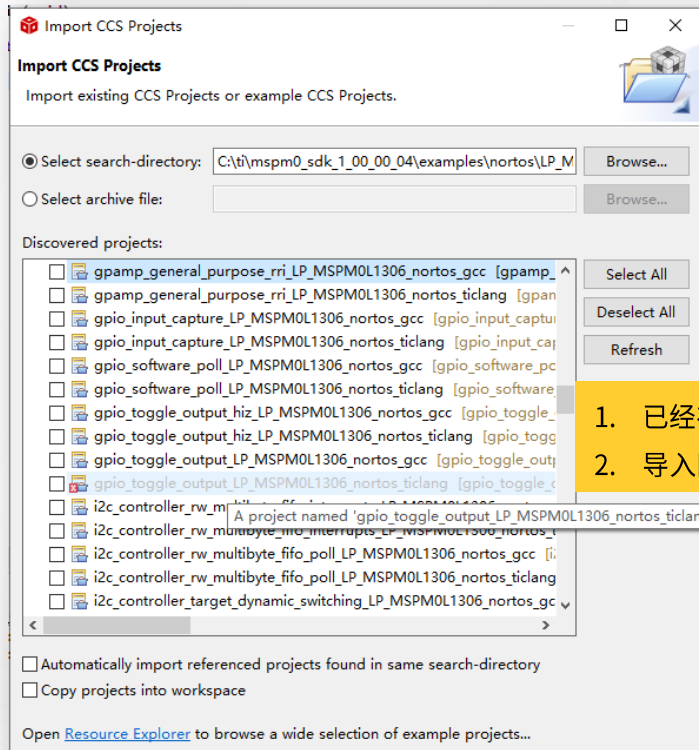


导入工程示例

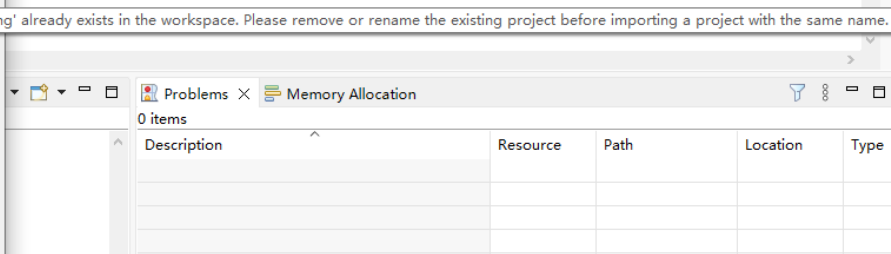




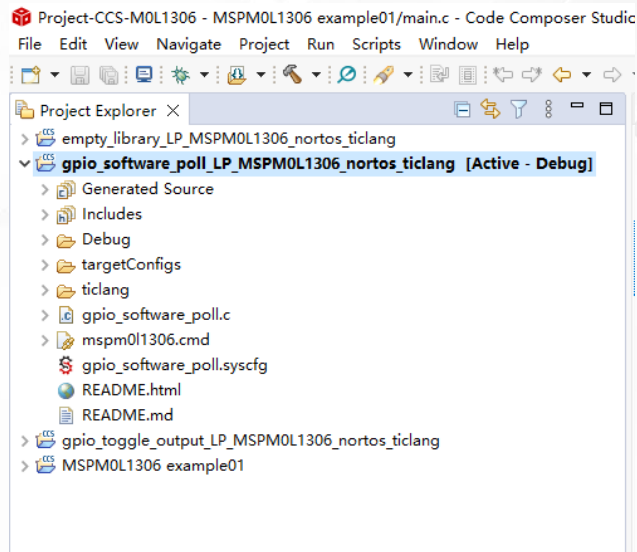
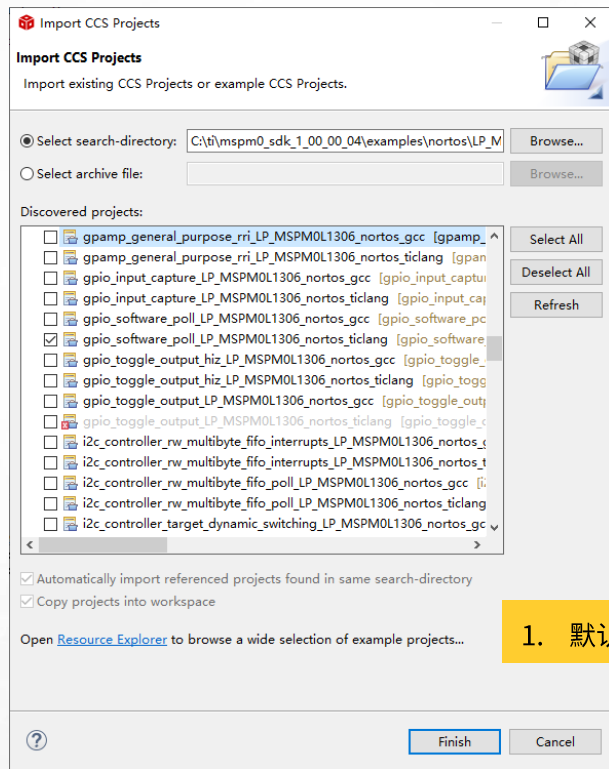
导入工程示例-已存在同名工程



1. 已经存在同名的工程，灰色显示
2. 导入同名工程，会提示出错！



导入工程示例 – 复制/链接



1. 默认会复制一份到工作空间下

导入工程示例

Project-CCS-M0L1306 - gpio_toggle_output_LP_MSPM0L1306_nortos_ticlang/README.md - Code Composer Studio

File Edit View Navigate Project Run Scripts Window Help

Project Explorer X

- ▼ gpio_toggle_output_LP_MSPM0L1306_nortos_ticlang [Active - D]
 - Generated Source
 - Includes
 - Debug
 - targetConfigs
 - ticlang
 - gpio_toggle_output.c
 - mbsp0l1306.cmd
 - gpio_toggle_output.syscfg
 - README.html
 - README.md

Getting Started Resource Explorer gpio_toggle_output.syscfg README.md X

Example Summary

Toggles three GPIO pins using HW toggle register.

Peripherals & Pin Assignments

Peripheral	Pin	Function
GPIOA	PA27	Standard Output
GPIOA	PA26	Standard Output
GPIOA	PA13	Standard Output

DEBUGSS PA19 Debug Data In Out

BoosterPacks, Board Resources & Jumper Settings

Visit [LP_MSPM0L1306](#) for LaunchPad information, including user guide and hardware files.

Pin	Peripheral	Function	LaunchPad Pin	LaunchPad Settings
PA27	GPIOA	PA27	J4_37	J13 ON/OFF: Connect/Disconnect LED2 Blue
PA26	GPIOA	PA26	J4_38	J12 ON/OFF: Connect/Disconnect LED2 Red
PA13	GPIOA	PA13	J4_39	J3 ON/OFF: Connect/Disconnect LED2 Green
PA20	DEBUGSS	SWCLK	J2_13	J101 15:16 ON: Connect to XDS-110 SWCLK (debug)

1. 导入成功的界面示意
2. 可以右键单击工程，在弹出的菜单中选择“Rebuild Project”

New >

Show In Alt+Shift+W >

Show in Local Terminal >

Add Files...

Copy Ctrl+C

Paste Ctrl+V

Delete Delete

Refactor >

Source >

Move...

Rename... F2

Import >

Export...

Show Build Settings...

Build Project

Clean Project

Rebuild Project

Refresh F5

Close Project

Build Targets >

Index >

Build Configurations >

Debug As >

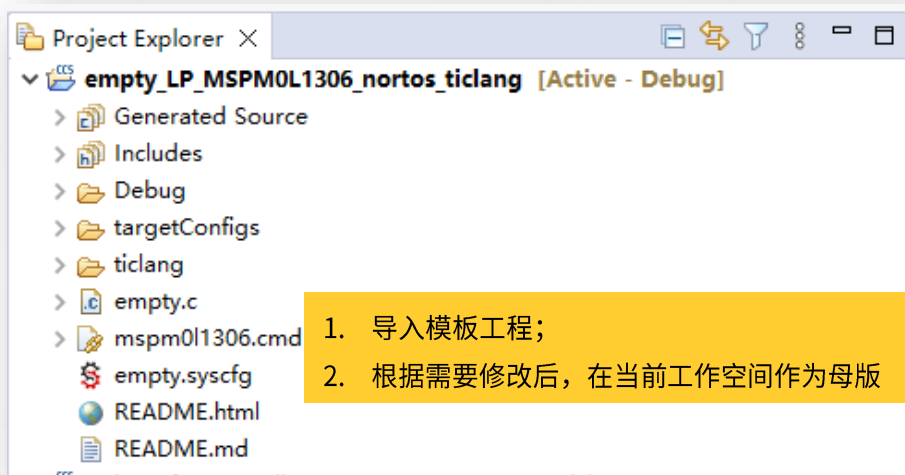
Restore from Local History...

Team >

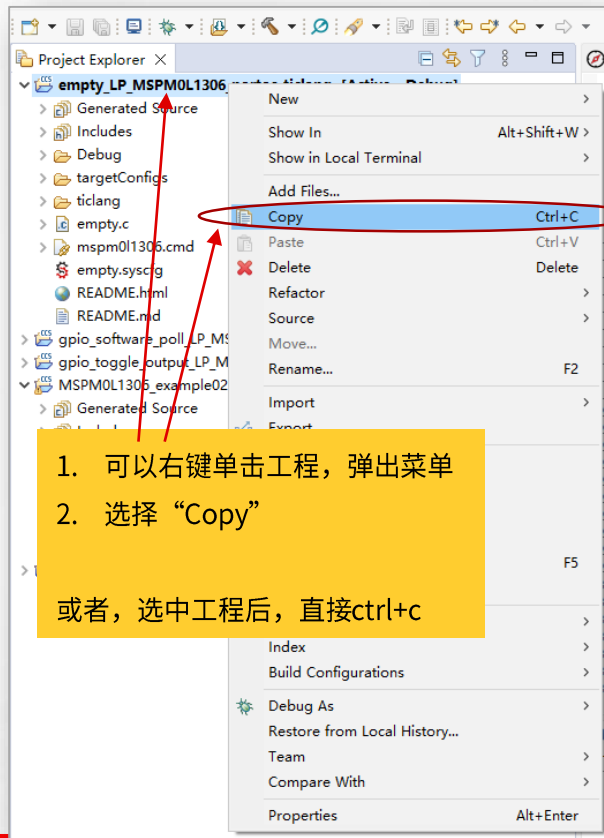
Compare With >

Properties Alt+Enter

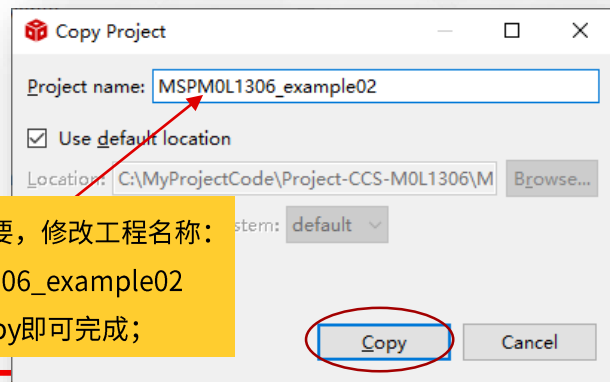
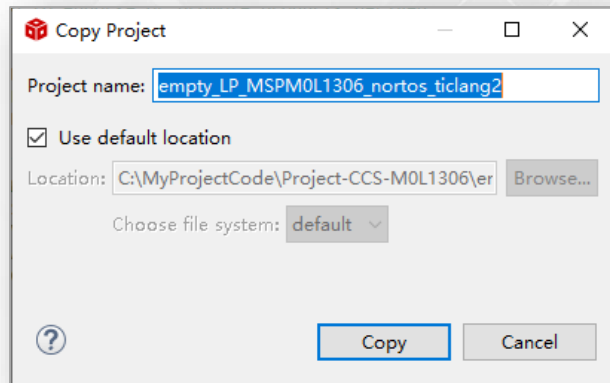
复制工程示例



复制工程示例



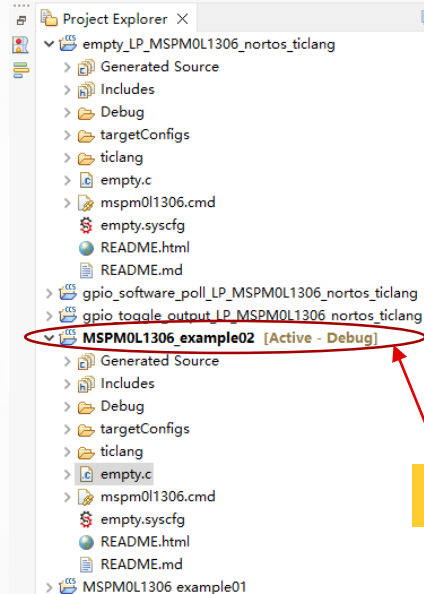
ctrl+v, 会弹出对话框



复制工程示例

Project-CCS-M0L1306 - MSPM0L1306_example02/empty.c - Code Composer Studio

File Edit View Navigate Project Run Scripts Window Help

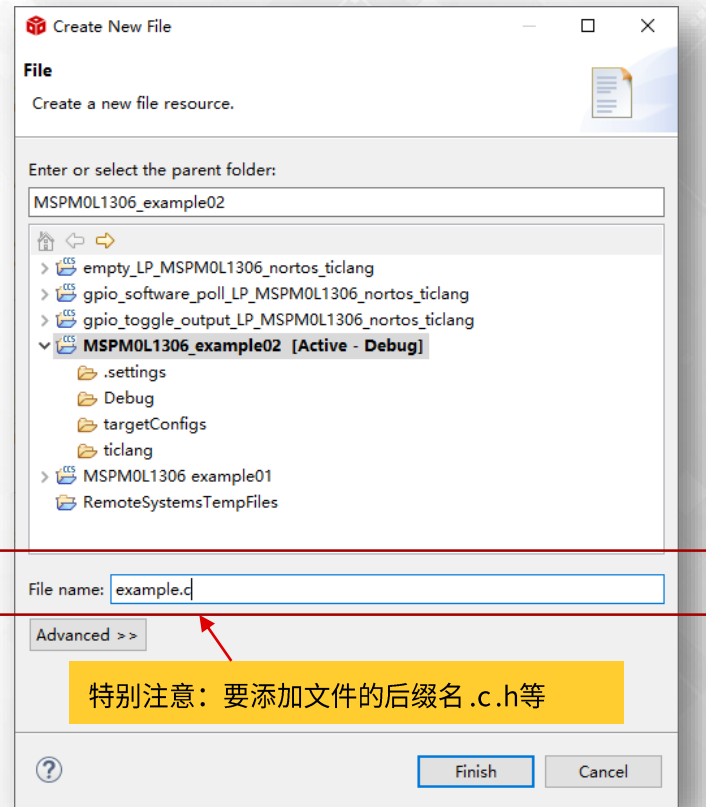
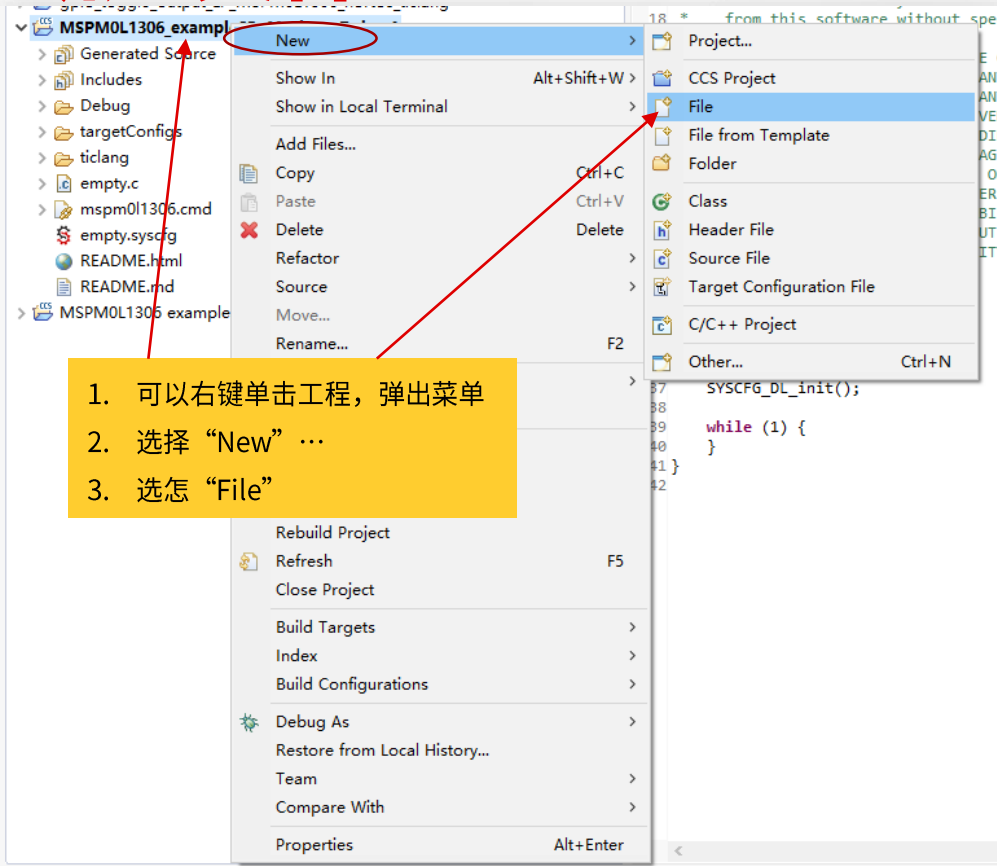


工程复制已完成!

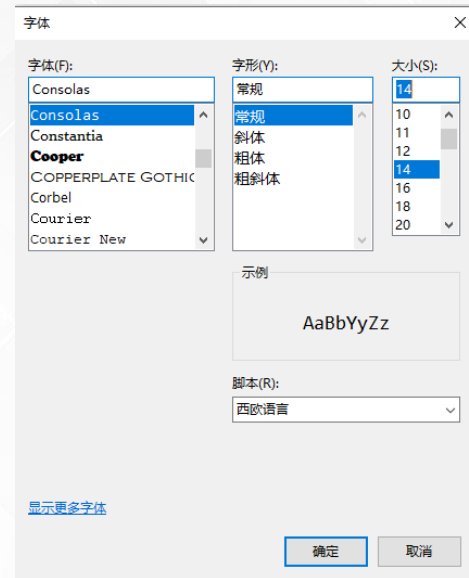
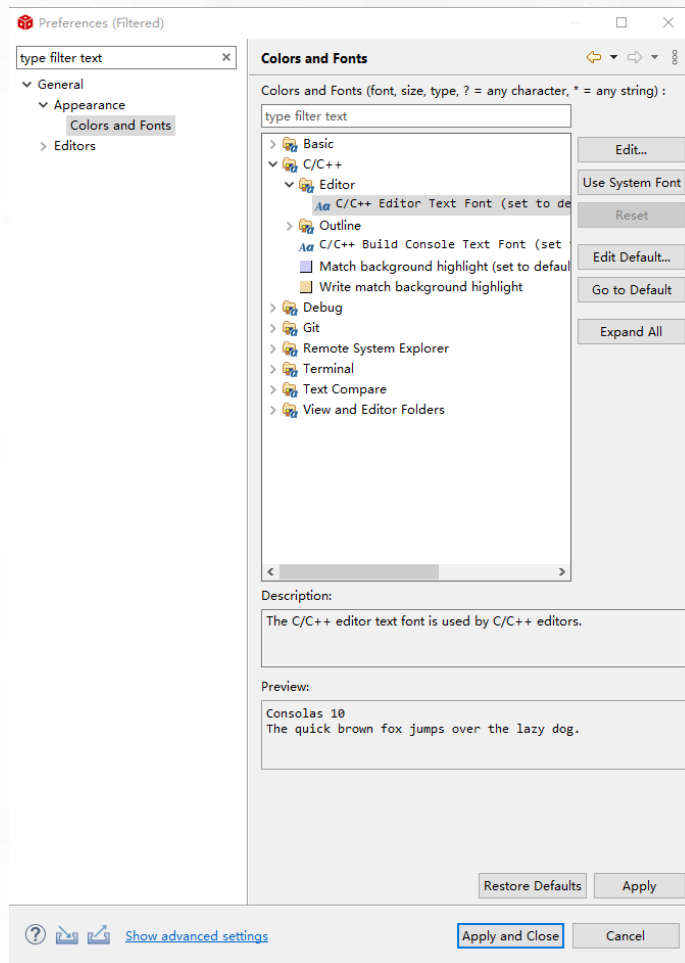
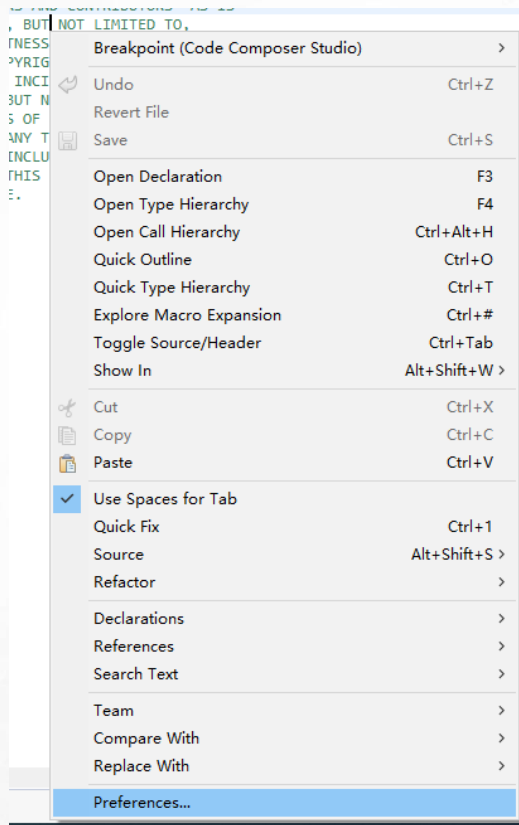
Resource Explorer empty.c x

```
1 /*
2  * Copyright (c) 2021, Texas Instruments Incorporated
3  * All rights reserved.
4  *
5  * Redistribution and use in source and binary forms, with or without
6  * modification, are permitted provided that the following conditions
7  * are met:
8  *
9  * Redistributions of source code must retain the above copyright
10 * notice, this list of conditions and the following disclaimer.
11 *
12 * Redistributions in binary form must reproduce the above copyright
13 * notice, this list of conditions and the following disclaimer in the
14 * documentation and/or other materials provided with the distribution.
15 *
16 * Neither the name of Texas Instruments Incorporated nor the names of
17 * its contributors may be used to endorse or promote products derived
18 * from this software without specific prior written permission.
19 *
20 * THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS"
21 * AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO,
22 * THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR
23 * PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR
24 * CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL,
25 * EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO,
26 * PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS;
27 * OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
28 * WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR
29 * OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE,
30 * EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
31 */
32
33 #include "ti_msp_dl_config.h"
34
35 int main(void)
36 {
37     SYSCFG_DL_init();
38
39     while (1) {
40     }
41 }
42
```

新建文件



字体的设置



实践环节03. SDK在CCS环境中的使用

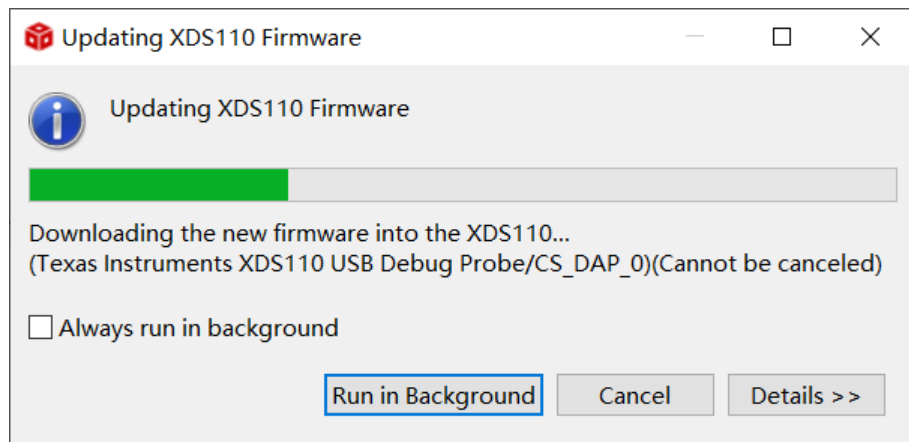
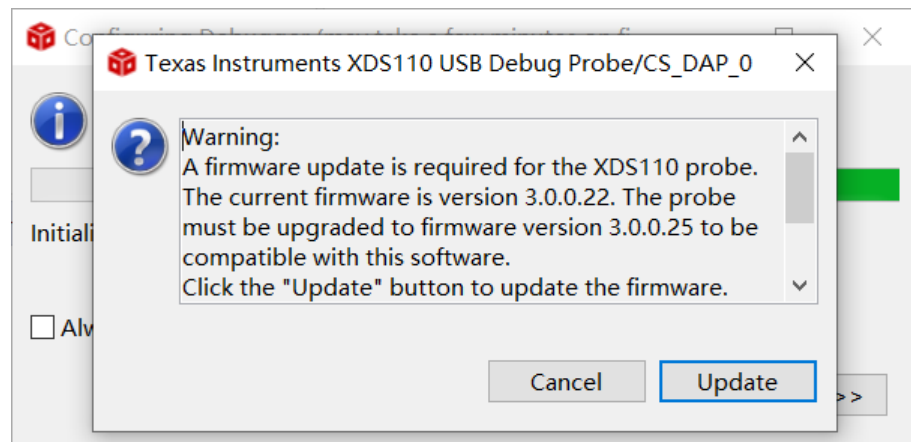
1. 导入SDK例程到CCS中

路径: C:\ti\mspm0_sdk_1_20_00_05\examples\nortos\LP_MSPM0G3507\driverlib

工程: gpio_toggle_output_hiz

2. 编译并构建工程
3. 连接Launchpad到电脑, 下载程序到Launchpad中

XDS110固件升级



谢谢!